Physical Data Model

DBMS: SQLite 3

SQLite was chosen for its simplicity: no server setup, a single portable `.db` file, native, and Python support.

SQL Schema

Tables

```sql
CREATE TABLE countries (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    name TEXT UNIQUE NOT NULL,
    iso_code TEXT,
    area_km2 REAL,
    capital TEXT,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE regions (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    name TEXT,
    nuts_code TEXT,
    population INTEGER,
    area_km2 REAL,
    tree_cover_km2 REAL,
    tree_cover_pct REAL
);

CREATE TABLE cities (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    name TEXT NOT NULL,
    country_id INTEGER,
    capital TEXT,
    fua_code TEXT,
    population INTEGER,
    urban_area_km2 REAL,
    lat REAL,
    lon REAL,
    eTRS_x REAL,
    eTRS_y REAL,
    source TEXT,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    urban_area_fua_km2 REAL,
    FOREIGN KEY (country_id) REFERENCES countries(id)
);

CREATE TABLE urban_trees (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    city_id INTEGER NOT NULL,
    source_year INTEGER,
    tree_cover_pixels BLOB,
    tree_cover_km2 REAL,
    urban_mask_pixels BLOB,
    urban_mask_km2 REAL,
    tree_percentage REAL,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
```

```sql
    FOREIGN KEY (city_id) REFERENCES cities(id)
);

CREATE TABLE urban_trees_ua (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    city_id INTEGER NOT NULL,
    urban_km2 REAL,
    forest_km2 REAL,
    green_km2 REAL,
    tree_pct REAL,
    rank INTEGER,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    FOREIGN KEY (city_id) REFERENCES cities(id)
);

CREATE TABLE city_rankings (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    city_id INTEGER NOT NULL,
    pop_density REAL,
    tree_density REAL,
    pop_to_tree_ratio REAL,
    rank INTEGER,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    FOREIGN KEY (city_id) REFERENCES cities(id)
);

CREATE TABLE population_annual (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    country_id INTEGER NOT NULL,
    year INTEGER NOT NULL,
    population_count INTEGER,
    FOREIGN KEY (country_id) REFERENCES countries(id),
    UNIQUE(country_id, year)
);

CREATE TABLE source_metadata (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    source_name TEXT NOT NULL,
    download_date DATE,
    record_count INTEGER,
    notes TEXT,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
```

Indexes

SQLite automatically creates an index for:

- Each `PRIMARY KEY` column (unique index)
- Each `UNIQUE` constraint

No additional indexes were explicitly defined due to the small data volume.

Constraints

Constraint Type	Example	Purpose
PRIMARY KEY	`countries.id`	Uniquely identifies each row
FOREIGN KEY	`cities.country_id → countries.id`	Referential integrity
NOT NULL	`cities.name`	Ensures required fields
UNIQUE	`countries.name`	Prevents duplicate entries
UNIQUE composite	`population_annual(country_id, year)`	One record per country per year
DEFAULT	`cities.created_at`	Automatic timestamp on insert

Data Types Used

SQLite Type	Usage	Rationale
INTEGER	IDs, populations, years, counts	Exact numeric values
REAL	Areas, densities, percentages, coordinates	Floating-point measurements
TEXT	Names, codes, descriptions, timestamps	String data
BLOB	Pixel data arrays	Raw numpy arrays stored as binary
DATE	Download dates	Date values in ISO format

Logical Data Model

Overview

The logical model takes the conceptual entities and turns them into a relational schema in Third Normal Form (3NF), breaking everything down to cut out redundancy while keeping data integrity intact using primary and foreign keys.

Normalization

1NF

All column values are atomic. Multi-valued attributes (like a country having multiple cities) are handled through separate tables with foreign keys instead of repeating groups.

2NF

Every table uses a single-column primary key, so partial dependencies aren't an issue. The `population_annual` table is a slight exception — it uses `(country_id, year)` as a unique constraint but keeps a surrogate `id` as the actual primary key.

3NF

All non-key attributes depend only on the primary key. For example, `cities.name` depends on `cities.id`, not on `country_id`. A couple of values like `urban_trees.tree_percentage` and `city_rankings.rank` are technically derivable but are stored for query performance.

Entity Relationship Diagram

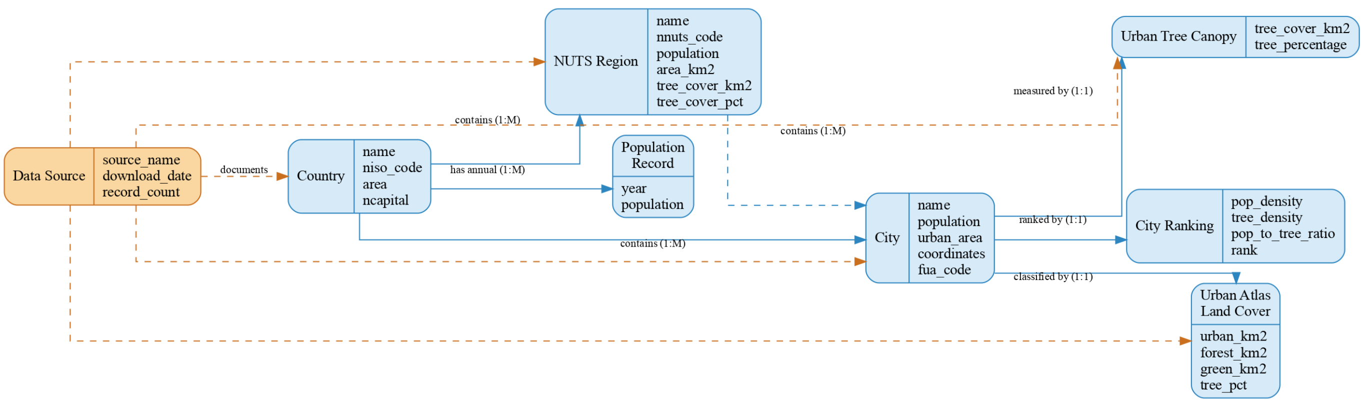

Table Definitions

countries

Column	Type	Constraints	Description
id	INTEGER	PK, AUTOINCREMENT	Unique country identifier
name	TEXT	NOT NULL, UNIQUE	Country name
iso_code	TEXT		ISO 3166-1 alpha-2 code
area_km2	REAL		Total land area in km²
capital	TEXT		Capital city name
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Row creation timestamp

regions

Column	Type	Constraints	Description
id	INTEGER	PK, AUTOINCREMENT	Unique region identifier
name	TEXT		Region/city name (may be NUTS code if unmapped)
nuts_code	TEXT		NUTS2 statistical code (Nomenclature of Territorial Units for Statistics)
population	INTEGER		Population from latest Eurostat year
area_km2	REAL		Functional Urban Area in km²
tree_cover_km2	REAL		Tree cover area in km²
tree_cover_pct	REAL		Tree cover as percentage of area

cities

Column	Type	Constraints	Description
id	INTEGER	PK, AUTOINCREMENT	Unique city identifier
name	TEXT	NOT NULL	City name
country_id	INTEGER	FK → countries.id	Owning country
capital	TEXT		Capital status description
fua_code	TEXT		Functional Urban Area code
population	INTEGER		City population
urban_area_km2	REAL		Urban area extent in km²
lat	REAL		Latitude (WGS84)
lon	REAL		Longitude (WGS84)
eTRS_x	REAL		ETRS89/LAEA easting
eTRS_y	REAL		ETRS89/LAEA northing
source	TEXT		Data source identifier
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Row creation timestamp
urban_area_fua_km2	REAL		FUA-derived urban area in km²

urban_trees

Column	Type	Constraints	Description
id	INTEGER	PK, AUTOINCREMENT	Unique measurement identifier
city_id	INTEGER	FK → cities.id, NOT NULL	Measured city
source_year	INTEGER		Year of the tree cover raster
tree_cover_pixels	BLOB		Raw pixel count of tree cover
tree_cover_km2	REAL		Tree cover area in km²
urban_mask_pixels	BLOB		Raw pixel count of urban mask
urban_mask_km2	REAL		Urban mask area in km²
tree_percentage	REAL		Tree cover as % of urban area
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Row creation timestamp

urban_trees_ua

Column	Type	Constraints	Description
id	INTEGER	PK, AUTOINCREMENT	Unique classification identifier
city_id	INTEGER	FK → cities.id, NOT NULL	Classified city
urban_km2	REAL		Urban land area in km²
forest_km2	REAL		Forest land area in km²
green_km2	REAL		Green urban area in km²
tree_pct	REAL		Tree cover percentage
rank	INTEGER		Rank by tree percentage
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Row creation timestamp

city_rankings

Column	Type	Constraints	Description
id	INTEGER	PK, AUTOINCREMENT	Unique ranking identifier
city_id	INTEGER	FK → cities.id, NOT NULL	Ranked city
pop_density	REAL		Population density (people/km²)
tree_density	REAL		Tree cover density (km² tree/km² urban)
pop_to_tree_ratio	REAL		Population per km² of tree cover
rank	INTEGER		Overall rank (1 = best tree equity)
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Row creation timestamp

population_annual

Column	Type	Constraints	Description
id	INTEGER	PK, AUTOINCREMENT	Unique record identifier
country_id	INTEGER	FK → countries.id, NOT NULL	Country
year	INTEGER	NOT NULL	Calendar year
population_count	INTEGER		Population for that year

source_metadata

Column	Type	Constraints	Description
id	INTEGER	PK, AUTOINCREMENT	Unique source identifier
source_name	TEXT	NOT NULL	Name of the data source
download_date	DATE		Date the data was downloaded
record_count	INTEGER		Number of records obtained
notes	TEXT		Additional documentation
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Row creation timestamp

Foreign Key Relationships

Source Table	FK Column	Target Table	Description
cities	country_id	countries	City belongs to a country
urban_trees	city_id	cities	Tree cover measurement belongs to a city
urban_trees_ua	city_id	cities	Land cover classification belongs to a city
city_rankings	city_id	cities	Ranking belongs to a city
population_annual	country_id	countries	Population record belongs to a country

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Conceptual Data Model

Overview

The conceptual model gives a high-level overview of the entities and relationships involved in the European Urban Tree Cover analysis, covering how population centers, administrative regions, and environmental measurements all connect to each other.

Entities (8)

1. Country

A European nation. Countries contain NUTS regions and cities, and maintain annual population records.

Attributes: name, ISO code, area (km²), capital city

2. NUTS Region

A NUTS2 statistical region defined by Eurostat. These are the primary analytical units for tree cover comparison. Some regions are metropolitan (mapped to a city name) while others are non-metropolitan.

Attributes: name, NUTS code, population, area (km²), tree cover (km²), tree cover (%)

3. City

A major European urban center. Cities serve as the link between population data and environmental measurements.

Attributes: name, population, urban area (km²), coordinates (latitude/longitude + ETRS projection), FUA code

4. Urban Tree Canopy

Tree cover measurements derived from the Copernicus Tree Cover Density 2015 raster at 20m resolution. Represents actual tree canopy within the urban footprint.

Attributes: tree cover pixels, tree cover (km^2), urban mask pixels, urban mask (km^2), tree percentage, source year

5. Urban Atlas Land Cover

Land cover classification from the Copernicus Urban Atlas 2021. Provides broader context including forest and green urban areas.

Attributes: urban area (km^2), forest area (km^2), green urban area (km^2), tree percentage, rank

6. City Ranking

Derived metrics ranking cities by tree density and population-to-tree ratios.

Attributes: population density, tree density, population-to-tree ratio, rank

7. Population Record

Annual population counts for each country.

Attributes: year, population count

8. Data Source

Metadata documenting the origin of each dataset used in the project.

Attributes: source name, download date, record count, notes

Relationships

Relationship	Type	Description
Country → NUTS Region	1:M	A country contains multiple NUTS regions
Country → City	1:M	A country contains multiple cities
Country → Population Record	1:M	A country has annual population records
NUTS Region → City	1:M	A region contains multiple cities (conceptual)
City → Urban Tree Canopy	1:1	A city has one tree cover measurement
City → Urban Atlas Land Cover	1:1	A city has one land cover classification
City → City Ranking	1:1	A city has one ranking entry
Data Source → (all entities)	M:N	Each source documents multiple entities

Why This Model

This model works well because it:

1. Keeps administrative units (Country, NUTS Region) separate from physical measurements (Urban Tree Canopy, Urban Atlas Land Cover), making it easy to analyze data at different geographic levels.
2. Links population data (Population Record, City) to environmental data (tree cover, land cover), which is central to answering the research question about equity between population and tree cover.
3. Tracks where data comes from through the Data Source entity, keeping the analysis reproducible.
4. Supports hierarchical analysis, country to region to city, with consistent foreign key relationships throughout.

Dataset Summary

Case Study

How do European NUTS2 regions compare in urban tree cover relative to their population?

This project analyzes the relationship between population density and tree canopy cover across European metropolitan regions, using data from Copernicus (tree cover density, urban land cover), Eurostat (population), and Wikipedia (city data).

Dataset Structure

The database contains 8 tables with a total of 433 records across all entities.

Entities Overview

Entity	Table	Rows	Description
Country	`countries`	51	European countries with ISO codes and area
NUTS Region	`regions`	212	NUTS2 regions with population and tree cover
City	`cities`	30	Major European cities with coordinates
Urban Tree Canopy	`urban_trees`	30	Tree cover from Copernicus TCD 2015 raster
Urban Atlas Land Cover	`urban_trees_ua`	29	Land cover classification from Urban Atlas 2021
City Ranking	`city_rankings`	28	Rankings based on tree density metrics
Population Record	`population_annual`	51	Annual population counts by country
Data Source	`source_metadata`	2	Metadata about data sources

Key Attributes

- **NUTS Regions:** name, NUTS code, population, area (km²), tree cover (km² and %)
- **Cities:** name, population, urban area (km²), coordinates (lat/lon + ETRS projection)
- **Urban Tree Canopy:** tree cover pixels, tree cover km², tree percentage (from 20m resolution TCD raster)
- **Urban Atlas Land Cover:** urban area, forest area, green urban area, tree percentage
- **City Rankings:** population density, tree density, population-to-tree ratio, rank

Data Sources

Source	Description	Records
Eurostat TGS00096	NUTS population data (2014–2024)	227 regions
Copernicus Urban Atlas 2021	FUA boundaries in GeoPackage format	768 metropolitan areas
Copernicus TCD 2015	Tree Cover Density raster at 20m resolution	1 raster covering EU
NUTS2021	NUTS classification with metropolitan labels	229 entries
Wikipedia	European country and city data	51 countries, 30 cities